

Semi-quantitative semantics

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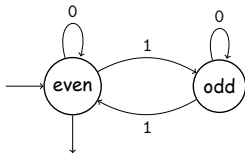
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Higher-order automata theory

There is a growing connection between

automata theory



and

λ -calculus

$\text{Church}_\Sigma \Rightarrow \text{Bool}$

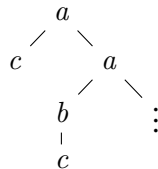
The main objects of study are

cartesian closed categories with **finite hom-sets**

Each one recognizes a class of higher-order languages, as defined by Salvati.

Qualitative methods in semantics

Record if a resource is used or not. Useful e.g. in higher-order model checking

$$\left\{ \begin{array}{l} S \quad := \quad F \ c \\ F \ x \quad := \quad a \ x \ (F \ (b \ x)) \end{array} \right. \rightsquigarrow$$


Qualitative tools are often **extensional** and compatible with **finite** structures.

Higher-order languages à la Salvati are well understood in that setting.

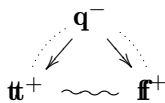
Quantitative methods in semantics

$$M = \sum_{n=0}^{\infty} \frac{D^n M}{n!}$$

Taylor expansion
and approximants

$$\frac{\Gamma_0 \vdash M : [q_1, \dots, q_n] \multimap q \quad \Gamma_1 \vdash N : q_1 \quad \dots \quad \Gamma_n \vdash N : q_n}{\Gamma_0 + \dots + \Gamma_n \vdash MN : q}$$

Non-idempotent
intersection types



Interactive
semantics

Quantitative tools are of an **intensional** and **infinite** nature.

How to connect intensional observations
with finite structures underlying higher-order automata?

Semi-quantitative semantics

We change the semiring of interest: ¹

$$\mathbb{N} = \{0, 1, 2, 3, \dots\} \rightsquigarrow \mathbb{S} = \{0, 1, 2+\}$$

which contains a common value $2+$ for all numbers greater or equal to 2.

The main idea is to identify multisets as follows:

$$[x, y, y] \sim [x, y, y, y] \sim [x, y, y, y, y] \sim \dots$$

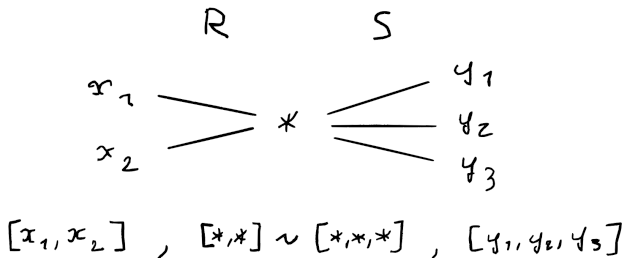
¹ Related to, but different from the work on multiplicity semirings by Carraro, Ehrhard, and Salibra, which focused on moving from $\mathbb{N}$ to $\mathbb{N} \cup \{\omega\} = \{0, 1, 2, 3, \dots, \omega\}$.

To choose a model

Every relation $R \subseteq X \times Y$ induces a relation between multisets

$$\left\{ ([x_1, \dots, x_n], [y_1, \dots, y_n]) : (x_i, y_i) \in R \text{ for } 1 \leq i \leq n \right\}$$

Not functorial in both **Rel** and **ScottL** (= Boolean profunctors).

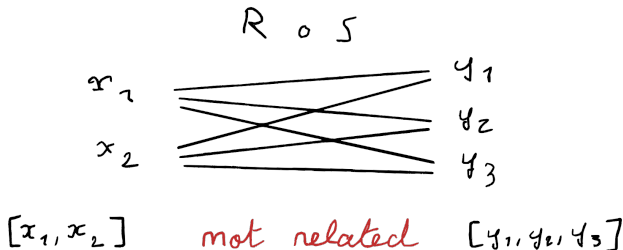


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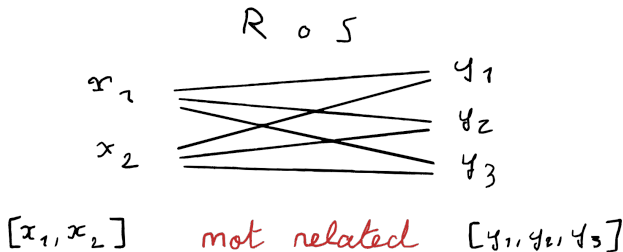


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► We move to coherence spaces.

Semi-multicliques

A **semi-multiclique** on the coherence space X is a function

$$u \quad : \quad |X| \longrightarrow \mathbb{S} = \{0, 1, 2+\}$$

whose support $\{x \in |X| \mid u(x) \neq 0\}$ is a clique of X .

The coherence space $!_{\mathbb{S}}X$ is defined as

$$\begin{aligned} |!_{\mathbb{S}}X| &:= \{\text{finite semi-multicliques}\} \\ \subset_{!_{\mathbb{S}}X} &:= \text{coherence of support cliques} \end{aligned}$$

Theorem. The exponential $!_{\mathbb{S}}$ is a Seely comonad on coherence spaces.

Intensionality

In the coKleisli category, global elements

$$u : !_{\mathbb{S}} \emptyset \cong 1 \longrightarrow X$$

are exactly cliques of X — not semi-multicliques.

As a consequence, a morphism of coherence spaces

$$R : !_{\mathbb{S}} X \longrightarrow Y$$

merely induces a set-theoretic function

$$\{\text{cliques of } X\} \longrightarrow \{\text{cliques of } Y\}$$

Fact. The semi-quantitative model is not well-pointed.

Remarkable denotational properties

	cliques	multicliques	semi-multicliques
Intensional?	×	✓	✓
Finitary?	✓	×	✓

Towards coherent multiplicity semirings

The semiring $\mathbb{S}$ is

- ▶ positive: if $x + y = 0$, then $x = 0$ and $y = 0$
- ▶ integral: $0 \neq 1$, and if $xy = 0$, then $x = 0$ or $y = 0$

Equivalently, the set-theoretic function

$$\begin{array}{ccc} \mathbb{S} & \longrightarrow & \mathbb{B} \\ x & \longmapsto & x \neq 0 \end{array}$$

into the Boolean semiring, is a homomorphism.

Conclusion

Future work:

- ▶ Intensional stable computation
- ▶ Intersection types and polyadic approximants
- ▶ Recognition of higher-order languages
- ▶ Unsafe higher-order tree transducers $\rightsquigarrow$ ask Tito!
- ▶ Extensional collapse on the qualitative model
- ▶ Taylor semi-expansion, and links with differentiation

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Thank you for your attention!

Any questions?